SMART INDIA HACKATHON 2026

- **Problem Statement Title:** Multi-Agent AI-Powered Ransomware Early Detection, Containment & Data Recovery System
- **PS Category :** Software
- **Team Name:** Cipher Syndicate

OUR TEAM

- **Vanshika Negi**
  Team lead
- **Priyanshi Saini**
- **Ansh Dhawan**
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- **Prakhar Srivastava**
- **Palak Saini**

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IDEA TITLE

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TRI-NETRA: Bharat's Next Generation Cyber Resilience Platform
{ Protect : Preserve : Prevail }

PROPOSED SOLUTION: A layered multi-agent ransomware defence and file recovery software for real-time file protection

PROBLEM

- Rapid file encryption causes immediate data loss
- Recovery is often complex, expensive, and time-consuming

INNOVATION AND UNIQUENESS

AI Powered Multi-Layer Detection

Combines behavioral analysis, file entropy checks & anomaly detection for high accuracy

Real time Threat Containment

Instant isolation of affected systems to prevent lateral movement.

Smart Recovery Mechanism

automated verified recovery using secure backups using minimal downtime

Adaptive & Self Learning Systems

continuously learns from new threats and evolves to stay ahead of attackers

proactive threat intelligence

Integrates global threat intel feeds to detect emerging ransomware variants early

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TECHNICAL APPROACH

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ATTACK
Ransomware attempt

GATEKEEPER
URL & File screening
- DNS Filtering
- Phishing Detection

WATCHDOG
File monitoring
- Entropy Analysis
- I/O Velocity

ML ANALYZER
Behaviour detection
- Risk Scoring
- Anomaly Detection

POLICY ENGINE
Threat decision
- Rule Engine
- Threat Validation

ENFORCER
Containment & recovery
- Process Kill
- File Locking

VAULTKEEPER
Protected backup
- AES-256
- Encrypted Backup

TECH STACK

LANGUAGES
Rust, Python

ML/DL FRAMEWORKS
Tensorflow, Pytorch, scikit-learn

DATA PROCESSING
Apache Kafka, Spark

DATABASE
PostGreSQL, Elasticseach

SECURITY
TLS ,AES-256, RBAC

DEPLOYMENT
Docker, Kubernetes CI/CD

1. CANARY FILES THAT FIGHT BACK
2. REMEMBRALL API INTERGATED FOR FILE RECOVERY
3. RECONSTRUCTS THE ENTIRE ATTACK SEQUENCE AFTER THE BREACH
4. LIVE RISK GRAPH AND EXPLAINABLE AI RISK SCORE

Detect → Defend → Recover

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FEASIBILITY & VIABILITY

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TECHNICAL FEASIBILITY

WINDOWS ENDPOINTS

System APIs + event logs

EXISTING TELEMETRY

Windows logs + network flows

SECURE RECOVERY

Versioned backups + integrity checks

LOCAL ML INFERENCE

ONNX-based on-device inferenc

INTEGRITY

Hashing + digital signatures

KEY RISKS

1. **FALSE POSITIVES**
2. **DETECTION DELAY**
3. **SLOW ATTACKS**
4. **BACKUP COMPROMISE**
5. **RESOURCE OVERHEAD**

MITIGATION

FALSE POSITIVES

↓

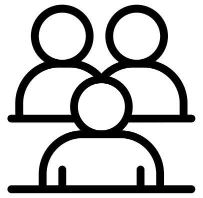
DETECTION DELAY

↓

BACKUP COMPROMISE

↓

RESOURCE OVERHEAD



Individuals

TARGET AUDIENCE

Small Business

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IMPACT & BENEFITS

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DATA COLLECTION

1. System activity
2. Process behaviour
3. Network traffic
4. File/Registry Changes

ANALYSIS

1. Multi-signal correlation
2. Behaviour analysis
3. ML inference
4. Threat classifications

THREAT DETECTION

Threat Detected?

YES

CONTINUOUS MONITORING

NO

CONTAINMENT

1. Isolate endpoint
2. Stop malicious processes
3. Quarantine files
4. Block connections

RECOVERY

1. Restore clean state
2. Verify integrity
3. Restart safely
4. Validate recovery

COMPETITORS & HOW WE ARE BETTER

Capability	Our Solution	CrowdStrike	Carbon Black	Microsoft	Sophos
AI behaviour detection	✓	✓	✓	✓	✓
Real-time containment	✓	✓	Partial	✓	Partial
Immutable recovery	✓	External	Limited	Limited	Limited
Local ML	✓	✓	✓	Limited	✓
Automated recovery	✓	Partial	Limited	Limited	Partial

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RESEARCH INSIGHTS

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RANSOMWARE BEHAVIOR

Behavioral + ML detection can identify ransomware activity beyond traditional signature-based approaches.

REAL-TIME CONTAINMENT

Modern EDR/XDR solutions use behavioral signals + automated isolation to limit ransomware spread.

TRUSTED RECOVERY

Protected and isolated backups help prevent attackers from compromising recovery data.

BEHAVIORAL ROLLBACK

Behavior-based ransomware protection can be combined with automatic file recovery/rollback.

KEY REFERENCES

CISA

#StopRansomware Guide

1. Backup
2. Isolation
3. Recovery
4. Resilience

Microsoft

#Defender for Endpoint & attack Disruption

1. AI protection
2. Signal correlation
3. Automated containment

Sophos

#CryptoGuard / Ransomware Protection

1. Behavioral detection
2. Rollback
3. Recovery

Broadcom

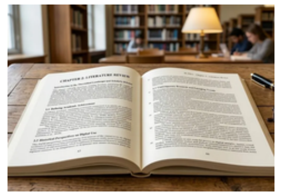
#Carbon Black Threat Prevention

1. Endpoint prevention
2. Behavioral analytics
3. Threat hunting

Research Paper

#Ransomware Detection & Classification Strategies

1. ML
2. Host-based detection
3. Network-based detection



Research Paper (2026)

#RansomTrack Hybrid Behavioral Analysis

1. Low-latency detection
2. Behavioral analysis
3. Explainable ML